

Final prediction in HMCN-F

$$P_F = \beta \cdot \underbrace{\left[P_L^1 \parallel P_L^2 \parallel \dots \parallel P_L^{|H|} \right]}_{\text{local}} + (1 - \beta) \cdot \underbrace{P_G}_{\text{global}}$$

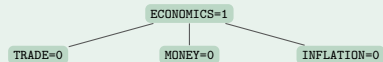
All metrics across β values

β	Fine labels (138)			Metacategories (12)			
	ROC	PR	F1	ROC	PR	BalAcc	F1
0.1	.806	.212	.210	.799	.605	.704	.576
0.3	.805	.211	.202	.799	.606	.706	.580
0.5	.802	.207	.196	.799	.605	.710	.580
0.7	.736	.147	.141	.799	.599	.699	.566
0.9	.717	.133	.131	.798	.600	.699	.568
1.0	.702	.125	.123	.797	.599	.693	.564
Range across all β				0.002	0.007	0.017	0.016

What HMCN-F Requires vs. Our Dataset

Reuters / FunCat (paper's datasets)

Every hierarchy node is **independently annotated**:

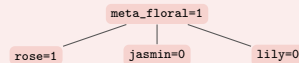


Parent=1 with all children=0 is **valid and frequent**: an article about economics with no specific sub-topic identified. Parent carries **independent information**.

Our dataset

Only **138 fine labels** are annotated by perfumers. Metacategories are **synthetic**:

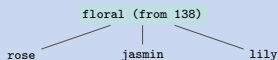
$$y_{\text{meta_floral}} = \max_{c \in \text{children}} y_c$$



Parent=1 with all children=0 is **impossible by construction**. Metacategory carries **zero independent signal** beyond its children.

Proposed restructuring

Use broad fine labels (floral, woody, fruity...) as natural parents *within the 138* independently annotated by perfumers, so parent=1 with children=0 might occur:



Two reasons it fails

Four metacategories have no valid parent node in the 138:

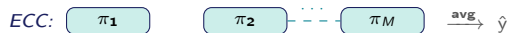
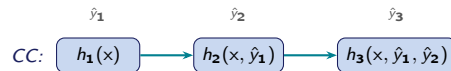
Metacategory	Problem
chemical	no node exists in 138
gourmand	no node exists in 138
animal_musk	two partial parents (animal + musk)
powdery_amber	two partial parents (powdery + amber)

Classifier Chain (CC)

Labels predicted sequentially; each classifier sees the input *and* previous predictions:

$$\hat{y}_j = h_j(x, \hat{y}_1, \dots, \hat{y}_{j-1})$$

One fixed ordering π over the L labels.



Ensemble of Classifier Chains (ECC)

Average over M chains, each with a *random* ordering π_m :

$$\hat{y}_j = \frac{1}{M} \sum_{m=1}^M h_j^{(\pi_m)}(x, \hat{y}_{<j}^{(\pi_m)})$$

Reduces variance from ordering choice.

Results : Metacategories (12 labels)

Model	ROC	PR	Bal.Acc	F1	Sens	Spec	ExM	Inst F1
BR+LR	0.802	0.683	0.693	0.624	0.709	0.677	0.037	0.654
BR+RF	0.804	0.697	0.694	0.620	0.644	0.744	0.031	0.655
CC_RF	0.488	0.260	0.503	0.189	0.215	0.790	0.004	0.170
ECC_RF	0.545	0.293	0.515	0.150	0.106	0.924	0.002	0.100

Layer	Role	Count	Examples
Layer 1	Root families	17	Floral, Fruity, Citrus, Spicy, Woody...
Layer 2	Sub-families	63	Rosy, Juicy Fruits, Cool Spices, Leather...
Layer 3	Leaves	21	Agrestic, Smoky Woods, Ozonic...
Total	All nodes	101	—

Layer 1 roots (17)

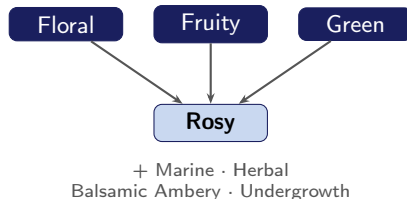
Floral · Fruity · Citrus · Herbal · Spicy · Woody · Green · Marine · Animalic · Musky · Undergrowth · Balsamic
Ambery · Burnt Leather · Aldehydes · Butyric Buttery · Sulfuric · Solvents

ScentTree is a DAG, not a Tree

A node can have *multiple* parents, reflecting genuine olfactory reality.

Example: **Rosy** is a descendant of **7** different Layer 1 families simultaneously.

If a fine label maps to *Rosy*, it sets **all** corresponding Layer 1 flags to 1 via OR-logic.



Category	Count	%
Exact name match	29	21.0%
Synonym mapping	34	24.6%
Total matched	63	45.7%
Absent from ScentTree	75	54.3%

Why 75 are absent

GoodScents/Leffingwell uses *concrete* descriptors (*apple, garlic, chocolate*).

ScentTree uses *abstract* family names (*Yellow Fruits, Sulfuric, Gourmand*).

All-paths

Label maps to Layer 1 node r if its ScentTree equivalent appears *anywhere* in subtree of r :

$$y_r = 1[\exists p \in \text{paths}(l) : r \in p]$$

2-hop

Label maps to r only if depth ≤ 2 :

$$y_r = 1[\text{depth}(l, r) \leq 2]$$

Metric	All-paths	2-hop
Avg. metacategories / label	3.83	2.86
Max (single label)	12	8
Labels with 1 parent	11	18
Labels with 2 parents	11	11
Labels with 3 parents	12	10
Labels with ≥ 4 parents	29	22
Labels with 0 parents	0	2*

* *powdery* and *waxy* only appear at depth 3+ under all roots.

Original 12

floral · fruity · sweet · woody
green · spicy · animal_musk
earthy · citrus · chemical
gourmand · powdery_amber

ScentTree 17

Node	vs. original
Floral, Fruity, Citrus	≈ same
Spicy, Woody, Green	≈ same
Herbal	split from green/spicy
Undergrowth	≈ earthy
Balsamic Ambery	≈ powdery_amber
Animalic + Musky	split from animal_musk
Marine	new
Burnt Leather	new
Butyric Buttery	new
Aldehydes	split from chemical
Sulfuric	split from chemical
Solvents	split from chemical

Proposed Mapping for Absent Labels : Fruits

Fine label	Closest ScentTree node	Proposed Layer 1	Conf.
apple	Yellow Fruits / Juicy Fruits	Fruity	Strong
apricot	Yellow Fruits	Fruity	Strong
banana	Yellow Fruits / Tropical Fruits	Fruity	Strong
peach	Yellow Fruits / Juicy Fruits	Fruity	Strong
pear	Green Fruits / Yellow Fruits	Fruity	Strong
plum	Yellow Fruits / Berries	Fruity	Strong
cherry	Berries / Juicy Fruits	Fruity	Strong
grape	Berries / Juicy Fruits	Fruity	Strong
raspberry	Berries	Fruity, Sulfuric	Strong
strawberry	Berries / Juicy Fruits	Fruity	Strong
black currant	Berries	Fruity, Sulfuric	Strong
pineapple	Tropical Fruits	Fruity, Sulfuric	Strong
melon	Juicy Fruits / Green Fruits	Fruity, Green	Strong
juicy	Juicy Fruits	Fruity, Green	Strong
ripe	Juicy Fruits / Yellow Fruits	Fruity	Medium
fruit skin	Green Fruits / Zesty	Fruity, Citrus	Medium

Proposed Mapping for Absent Labels : Gourmand & Boozy

Gourmand

Label	Proposed Layer 1	Conf.
caramellic	Balsamic Ambery	Strong
chocolate	Balsamic Ambery, Burnt Leather	Strong
cocoa	Balsamic Ambery, Burnt Leather	Strong
coffee	Balsamic Ambery, Burnt Leather	Strong
hazelnut	Balsamic Ambery, Fruity, Woody	Strong
popcorn	Balsamic Ambery, Burnt Leather	Medium
sweet	Balsamic Ambery	Medium
creamy	Fruity, Woody	Medium
dairy	Butyric Buttery, Animalic	Medium

Boozy / Fermented

Label	Proposed Layer 1	Conf.
alcoholic	Fruity, Balsamic Ambery	Strong
brandy	Balsamic Ambery, Fruity	Strong
cognac	Balsamic Ambery, Fruity	Strong
rummy	Balsamic Ambery, Fruity	Strong
winey	Fruity, Balsamic Ambery	Strong
fermented	Animalic, Balsamic Ambery	Medium
malty	Balsamic Ambery, Burnt Leather	Medium

Animal / Butyric

Label	Proposed Layer 1	Conf.
cheesy	Butyric Buttery, Animalic	Strong
beefy	Animalic, Butyric Buttery	Medium
meaty	Animalic, Burnt Leather	Medium
fishy	Animalic, Marine	Medium
sweaty	Animalic, Butyric Buttery	Medium

Savory / Spicy

Label	Proposed Layer 1	Conf.
clove	Spicy, Balsamic Ambery	Strong
chamomile	Herbal, Floral	Strong
cooling	Herbal, Spicy	Medium
cortex	Spicy, Balsamic Ambery	Medium
savory	Spicy, Herbal	Medium
cooked	Burnt Leather, Balsamic Ambery	Weak
warm	Spicy, Woody, Balsamic Ambery	Weak
aromatic	Herbal, Spicy	Weak

Proposed Mapping for Absent Labels : Green & Vegetal

Fine label	Closest ScentTree node	Proposed Layer 1	Conf.
leafy	Cut Grass / Crisp Green	Green	Strong
hay	Cut Grass / Agrestic	Green, Herbal, Undergrowth	Strong
weedy	Agrestic / Cut Grass	Green, Undergrowth	Strong
garlic	Sulfuric	Sulfuric, Green	Strong
onion	Sulfuric	Sulfuric, Green	Strong
alliaceous	Sulfuric	Sulfuric	Strong
cucumber	Aquatic / Crisp Green	Green, Marine	Medium
tomato	Crisp Green / Agrestic	Green, Herbal	Medium
celery	Agrestic / Green	Green, Herbal	Medium
cabbage	Agrestic / Sulfuric	Green, Sulfuric	Medium
radish	Sulfuric / Agrestic	Sulfuric, Green	Medium
vegetable	Agrestic / Crisp Green	Green, Undergrowth	Medium
potato	Agrestic / Earthy	Undergrowth, Green	Medium

Proposed Mapping for Absent Labels : Chemical, Floral & Misc

Chemical / Solvent

Label	Proposed Layer 1	Conf.
ethereal	Solvents, Fruity	Strong
ketonic	Solvents	Medium
phenolic	Spicy, Burnt Leather	Medium
soapy	Aldehydes, Floral	Medium
gassy	Solvents, Sulfuric	Medium
pungent	Sulfuric, Solvents	Medium
sharp	Solvents, Sulfuric	Weak
clean	Marine, Aldehydes	Weak
dry	Woody, Undergrowth	Weak
bitter	Citrus, Herbal	Weak
sour	Citrus, Sulfuric	Weak
oily	Green, Woody	Weak

Floral

Label	Proposed Layer 1	Conf.
hyacinth	Floral	Strong
lily	Floral	Strong
muguet	Floral, Green	Strong
hawthorn	Floral, Herbal	Medium

Labels to drop

Label	Reason
natural	not an olfactory category
odorless	not an olfactory category